Sawyer Jungemann

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PROFESSIONAL SUMMARY

Mechanical engineering student (CSWA-certified in SolidWorks) with hands-on manufacturing engineering experience at an AS9100-certified aerospace supplier. Skilled in GD&T (ASME Y14.5), print interpretation, dimensional inspection, fixture design, and design for manufacturability, with production-floor experience root-causing nonconformances, executing corrective actions, and building processes that hold tight tolerances.

EDUCATION

Arizona State University — B.S. Mechanical Engineering **Tempe, AZ**
Transferring Spring 2027 *Expected Graduation Fall 2028*

Glendale Community College — A.S. Engineering, Honors, Phi Theta Kappa **Glendale, AZ**
GPA: 3.80/4.0 (Technical GPA: 3.90/4.0) | *Honors Program* | *Engineering Club* *Expected Dec 2026*
Relevant Coursework: Calculus I–III, Differential Equations, Physics I–II, Statics, Dynamics, Mechanics of Materials, Linear Algebra, MATLAB

EXPERIENCE

Mod-A-Can, Inc. — Manufacturing Engineering Intern **Phoenix, AZ**
AS9100-certified aerospace sub-tier supplier *May 2026 – Aug 2026*

- Root-caused a brazing nonconformance with the engineering team to a non-magnetic base material limiting induction heating to eddy currents alone; helped design a steel susceptor and revised setup that replaced an inconsistent manual torch process and brought parts within a ±0.004" flatness spec
- Assisted a junior engineer in developing a pin-based check gauge (Autodesk Inventor) to verify a ±1° angle across three holes where the Keyence lacked a rotary attachment; oversaw the machinist through fabrication
- Interpreted customer prints from Honeywell, Collins Aerospace, L3Harris, Moog, and U.S. government programs, applying GD&T (ASME Y14.5) to verify processes and part conformance and resolving design and machining questions on the shop floor
- Performed dimensional inspection across 40+ production jobs, including thin-wall avionics housings and brazed aluminum/beryllium copper assemblies, using MicroVu/Keyence optical systems and hand gauges
- Identified 10+ nonconformances and developed rework plans with engineering; flagged parts, signed off on NCR reports, and executed floor-level corrective actions (manual mill, air press, pin/bushing pressing, HELICOIL, gasket work)
- Applied Lean principles to redesign a 600+ tool locator system, clearing a major inventory backlog and cutting tool retrieval time ~80%
- Built and trained a company AI knowledge base spanning paint/adhesive specs, work instructions, SDS, and TDS documents, plus a material-compatibility cross-reference tool, cutting document search time from 30+ min to 3–5 min
- Created and owned a shop-floor system for recording job operations; trained operators to record independently, capturing 125+ operations and 8+ hours of footage for work instruction development

NASA ASCEND Program — Student Engineer, Team Captain **Aug 2026 – Present**

- Selected as one of 10 students school-wide for a NASA-funded program to design, build, and launch a payload to 100,000 ft
- Lead a multidisciplinary team integrating the payload's mechanical and electrical systems
- Designed a flight data recorder (FDR) enclosure for 10 g vibration/impact loads to protect avionics through balloon ascent and descent, across -85°F to 90°F
- Fabricated the FDR from PCB etching through component soldering; contributed to avionics code (C++)

Harumi Sushi & Sake — Server **Peoria, AZ**
Feb 2024 – May 2026

- Reconciled $5,000–$10,000 in nightly revenue with no cash discrepancies while coordinating team service at high volume

PROJECTS

Custom Fixed-Wing UAV — Airframe & Propeller Design

- Designed a fixed-wing airframe and custom propeller geometry in SolidWorks for thrust efficiency; parts currently in fabrication
- Built a custom thrust stand for static thrust measurement, with a commercial propeller as baseline

Balsa Wood Warren Truss Bridge — Design & Load Testing

- Modeled truss in SolidWorks with dimensioned drawings; led fabrication of the structure using shop tools
- Load-tested to destruction, reaching a 403 lb failure load on a sub-1-lb frame (403:1 load-to-weight ratio)

SKILLS

CAD/Design: SolidWorks (CSWA Certified), Autodesk Inventor, Fixture/Check Gauge Design, Engineering Drawings, Design for Manufacturability (DFM)

Quality & Manufacturing: GD&T (ASME Y14.5), Print Interpretation, Root-Cause Analysis, Dimensional Inspection, MicroVu/Keyence Optical Measurement, Micrometers/Calipers/Height Gauges, NCR Reporting, Corrective Action, Lean, Work Instruction Development

Software: MATLAB, JobBoss ERP, Excel, Arduino, C++, AI/LLM Knowledge Base Tools

Fabrication: Manual Mill, Air Press, Drill Press, Band Saw, Belt Sander, Mechanical Testing, Soldering, PCB Etching